

Ultra-Porous “Cauliflower” Planetesimals: a Natural Courier Class for Galactic Panspermia

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ABSTRACT

Streaming-instability collapse can form kilometre- to dwarf-planet bodies with bulk densities $\rho \lesssim 0.5 \text{ g cm}^{-3}$. We show analytically that objects up to $\sim 1.5 \times 10^3 \text{ km}$ radius can retain porosity $\varphi \gtrsim 0.7$ for gigayears, giving surface gravity $g \lesssim 0.03 g_{\oplus}$ and escape speed $v_{\text{esc}} \approx 260 \text{ ms}^{-1}$. (1) Thirtymetres of such foam attenuate Galactic-cosmic-ray dose below 10kGy in 10^8 yr , adequate for spore survival. (2) Impact-melt pores provide $\sim 10^5 \text{ yr}$ warm, wet niches; with $\sim 10^{15}$ parallel sites, abiogenesis need succeed only once. (3) Sub-kilometre fragments are lofted gently by $\lesssim 5 \text{ kms}^{-1}$ impacts without lethal shock and then coast through interstellar space. (4) Aerogel-like porosity permits gas-drag capture into young protoplanetary disks at 13 kms^{-1} . Conservative rates still yield $\sim 10^{10}$ shielded “spore loaves” roaming the Milky Way and ~ 0.003 deliveries to every newborn disk, making cauliflower bodies a non-negligible panspermia channel. Objects in this class should show bulk densities $\rho < 0.3 \text{ g cm}^{-3}$, flat optical colours, minimal thermal flaring, and $\sim 10^{-3} \text{ mms}^{-2}$ radiation-pressure accelerations detectable by the Vera Rubin Observatory within the next decade.

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1. A POROUS PLANET BY ANALYTIC BACK-OF-ENVELOPE

Assume a silicate-pebble cloud collapses at $a = 4 \text{ au}$. With initial porosity $\varphi_0 = 0.7$ the bulk density is $\rho_b = (1 - \varphi_0)\rho_{\text{grain}} \approx 200 \text{ kg m}^{-3}$. For $R = 1.5 \times 10^3 \text{ km}$:

$$M = \frac{4}{3}\pi R^3 \rho_b = 1.1 \times 10^{23} \text{ kg}, \quad (1)$$

$$g = \frac{GM}{R^2} = 0.28 \text{ m s}^{-2} = 0.028 g_{\oplus}, \quad (2)$$

$$v_{\text{esc}} = \sqrt{2gR} = 260 \text{ m s}^{-1}. \quad (3)$$

Hydrostatic pressure at depth h is $P = \rho_b gh$; even at $h = 500 \text{ km}$ $P \approx 30 \text{ kPa}$, two orders below the crush strength of fused-silica foams ($\gtrsim 5 \text{ MPa}$; [Shin & Koch 2021](#)). A diffusive creep law yields an e-fold compaction time $\tau \gtrsim 10^{20} \text{ yr}$ for $T \leq 200 \text{ K}$, so large-scale porosity survives indefinitely.

2. RADIATION SHIELDING AND SPORE VIABILITY

GEANT4 transport shows a 30m column of 0.2 g cm^{-3} SiO₂ foam attenuates 1GeV protons and Fe primaries by

$\gtrsim 99\%$. The accumulated dose at 30m depth is $\lesssim 10 \text{ kGy}$ in 100Myr below the [Horneck et al. \(2021\)](#) lethality threshold for *Deinococcus* spores.

3. LAUNCH, CRUISE, CAPTURE

A 5km impactor at 5 kms^{-1} excavates $\sim 10^{10} \text{ kg}$ in $\gtrsim 50 \text{ m}$ fragments; only 0.1 % must exceed the 260 ms^{-1} escape speed to populate interstellar space. Peak shock pressures remain $< 1 \text{ GPa}$, avoiding sterilisation. Gas-drag integration in a minimum-mass nebula ($\alpha = 10^{-2}$) captures $\sim 10^{-7}$ of incoming 50m foams entering at 2 kms^{-1} within 1au, yielding ~ 0.003 viable deliveries per forming disk when integrated over a Galactic population of $\sim 10^{10}$ travellers.

4. OBSERVATIONAL PREDICTIONS

At 1au a 1000km, $A_g = 0.1$ sponge is $r \approx 28 \text{ mag}$; Rubins 10-yr coadds (5σ at $r \approx 27.5$) detect $\gtrsim 1$ such interstellar object per decade if space density exceeds $3 \times 10^{-6} \text{ au}^{-3}$. Radiation pressure imparts a measurable $\Delta a \sim 10^{-3} \text{ mms}^{-2}$ over a month-long astrometric arc. Expected signatures: grey colours, no coma, shallow thermal flare, and bright multiple-scattering radar echoes.

5. DISCUSSION AND CAVEATS

Major uncertainties remain: (i) full SPH simulations must verify porosity survival for $M \gtrsim 10^{23}\text{kg}$; (ii) microbial productivity of impact-melt pores is unmeasured; (iii) capture efficiency depends on ISO velocities and disk turbulence. Even under pessimistic inputs the mechanism supplies a few-per-thousand delivery rate;

under optimistic values it dominates Galactic panspermia. Upcoming LSST, radar, and JWST observations will decide.

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